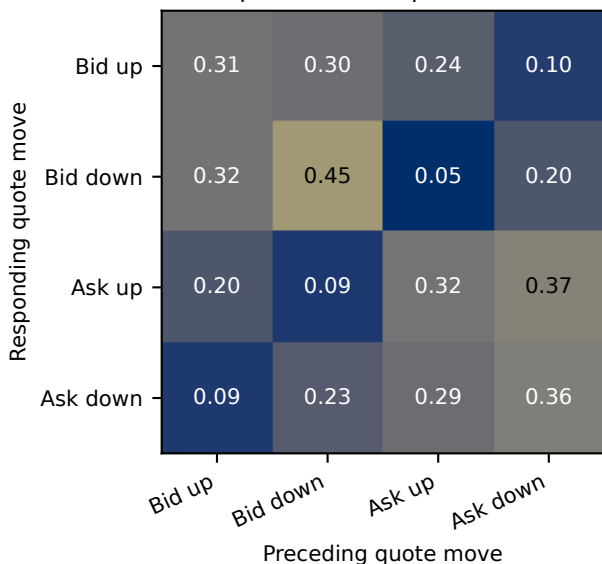
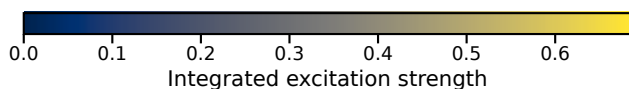
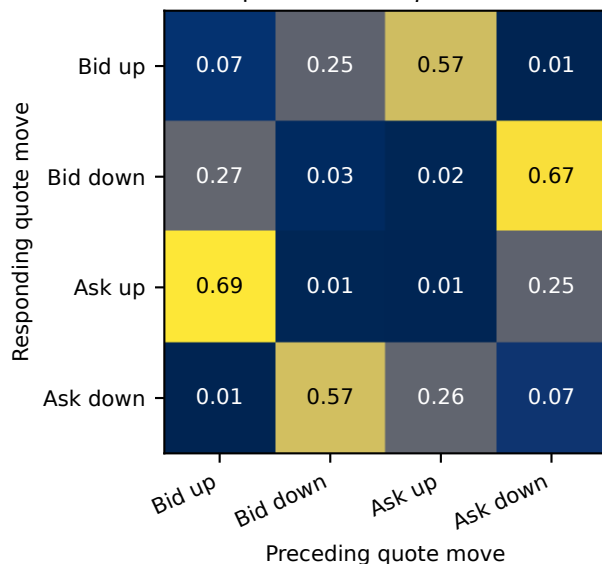


Hawkes excitation kernels for EUR/USD

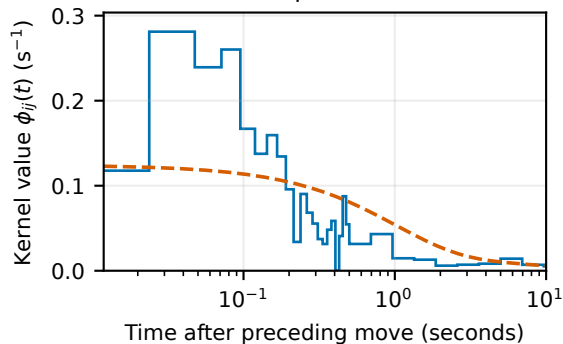
(a) Conditional least squares
Spectral radius $\rho = 0.9784$



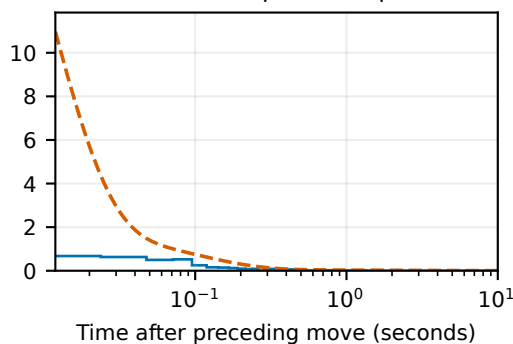
(b) Converged EM fit
Spectral radius $\rho = 0.9373$



(c) Bid up $\rightarrow$ Bid down



(d) Bid up $\rightarrow$ Ask up



— Conditional least squares - - - Converged EM fit